

Effects of tomato extract on platelet function: a double-blinded crossover study in healthy humans.



BACKGROUND: Aqueous extracts from tomatoes display a range of antiplatelet activities in vitro. We previously showed that the active components also alter ex vivo platelet function in persons with a high response to ADP agonist. OBJECTIVE: The objective was to evaluate the suitability of a tomato extract for use as a dietary supplement to prevent platelet activation. DESIGN: A randomized, double-blinded, placebo-controlled crossover study was conducted in 90 healthy human subjects selected for normal platelet function. Changes from baseline hemostatic function were measured 3 h after consumption of extract-enriched or control supplements. RESULTS: Significant reductions in ex vivo platelet aggregation induced by ADP and collagen were observed 3 h after supplementation with doses of tomato extract equivalent to 6 (6TE) and 2 (2TE) tomatoes [3 micromol ADP/L: 6TE (high dose), -21.3%; 2TE (low dose), -12.7%; $P < 0.001$; 7.5 micromol ADP/L: 6TE, -7.8%, 2TE, -7.6%; $P < 0.001$; 3 mg collagen/L: 6TE, -17.5%; 2TE, -14.6%; $P = 0.007$]. No significant effects were observed for control supplements. A dose response to tomato extract was found at low levels of platelet stimulation. Inhibition of platelet function was greatest in a subgroup with the highest plasma homocysteine ($P < 0.05$) and C-reactive protein concentrations ($P < 0.001$). CONCLUSION: As a functional food or dietary supplement, tomato extract may have a role in primary prevention of cardiovascular disease by reducing platelet activation, which could contribute to a reduction in thrombotic events.